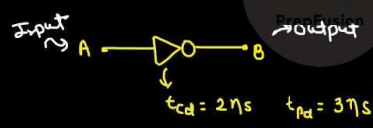
Static Timing Analysis

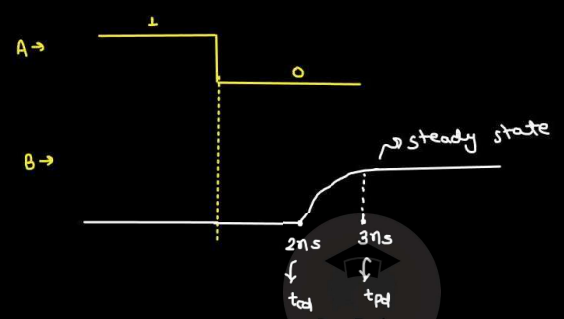
* Important terms:-

① Contamination delay and propagation delay (T_{cd} and T_{pd})



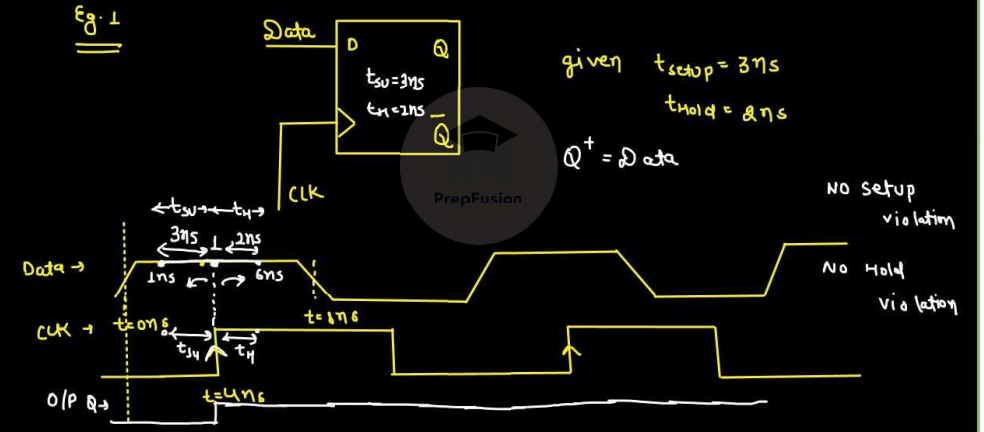
at t=0; A is changed from 1 to 0.

Draw waveform at B.



* Setup time and Hold time (t_s and t_H):-

Eg. 1



given $t_{setup} = 3ns$

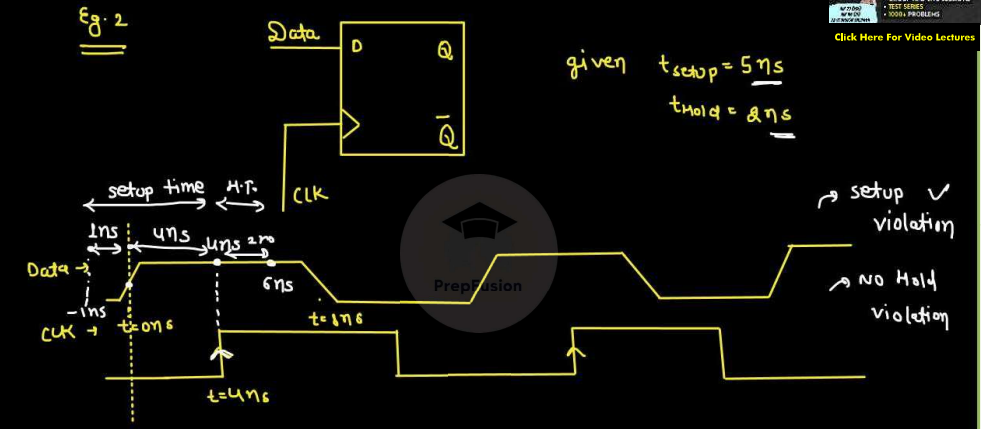
$t_{Hold} = 2ns$

$Q^+ = Data$

NO setup violation

No Hold violation

Eg. 2



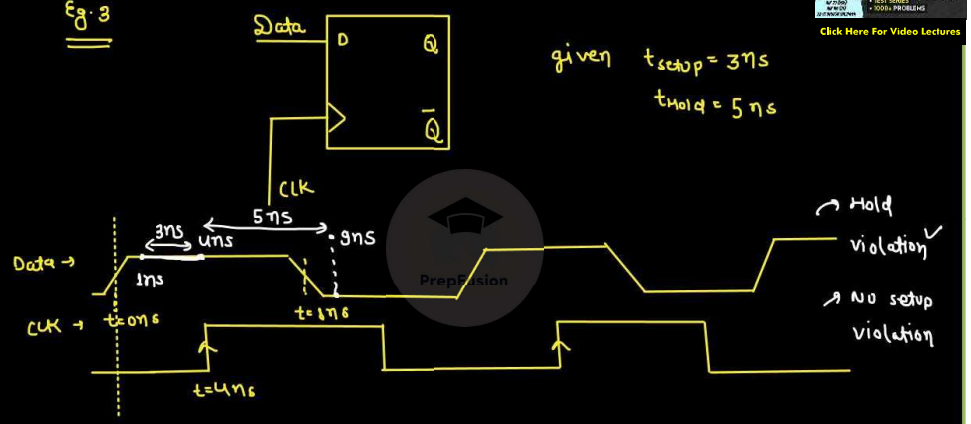
given $t_{setup} = 5ns$

$t_{Hold} = 2ns$

setup violation ✓

No Hold violation

Eg. 3



given $t_{setup} = 3ns$

$t_{Hold} = 5ns$

↗ Hold violation ✓

↗ No setup violation

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Definition:-

★ Setup time of a flip-flop:-

Amount of time the i/p must be stable before clock transition.

★ Hold time of a flip-flop:-

Amount of time the i/p must be stable after clock Transition.

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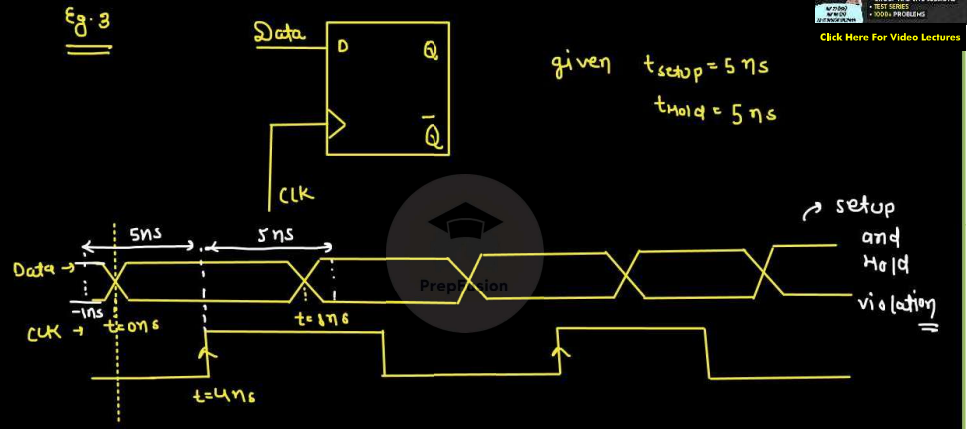
Eg. 3



given $t_{setup} = 5ns$

$t_{Hold} = 5ns$

↗ setup and Hold violation

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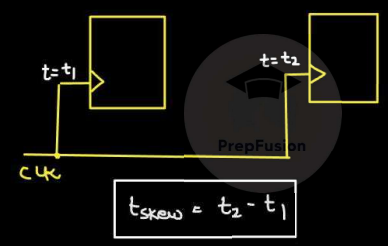
Clock skew:-

Difference in the arrival time of same clock at different points in the design.



$t_{skew} = t_2 - t_1$

[Assuming $t_2 > t_1$]

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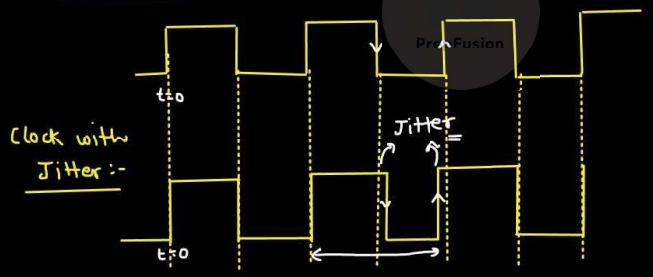
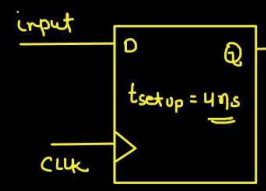
Clock Jitter:-

Natural periodicity of clock being disrupted because of the external source like noise, temp variation and voltage variations.

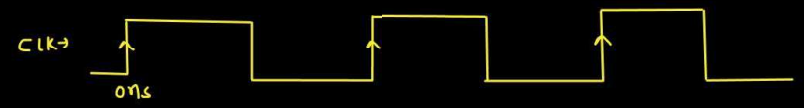
ideal clock:-



Q.



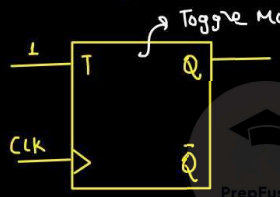
Input is changed from 0 to 1 at t = 3ns, and remains constant. For proper operation; what should be min clock Time period? what should be the maximum Hold time of the flip-flop?



$Q^+ = D$ = input

* Clock to Q delay of a flip flop:-

[t_{c-Q}] [Propagation delay]

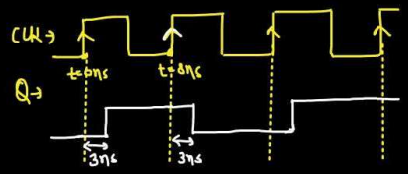


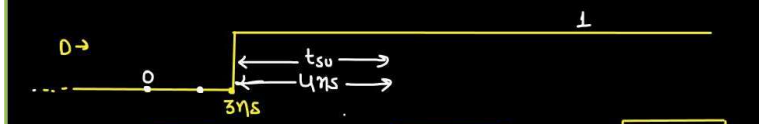
$Q^+ = \bar{Q}$

Take Q (initial) = 0

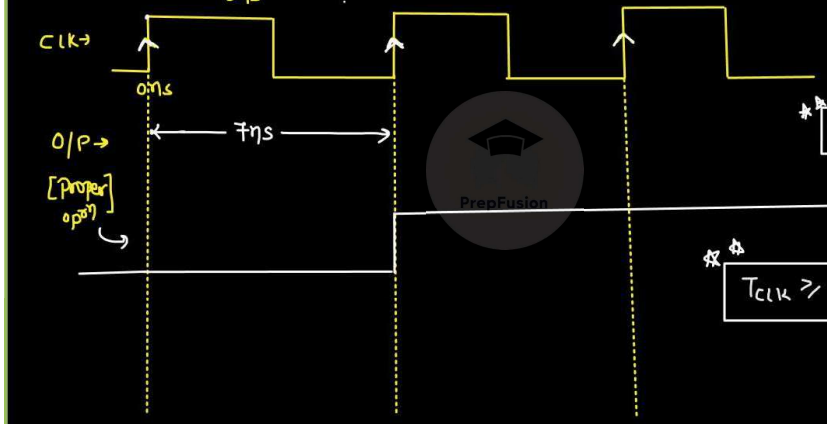
$t_{c-q} = 3ns = t_{pd}$

Draw waveform at Q.





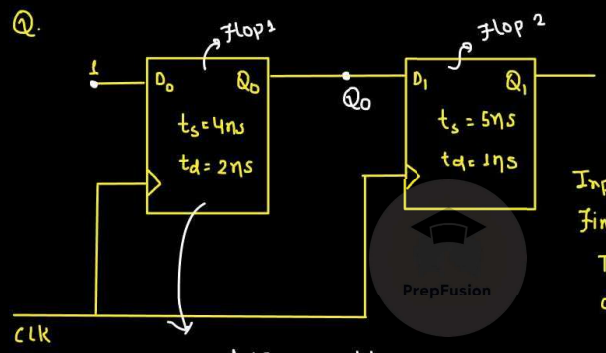
$t_{su} = 4ns$



* $t_H \leq 3ns$

* $T_{CLK} \geq 7ns$

Q.



no setup or Hold violation

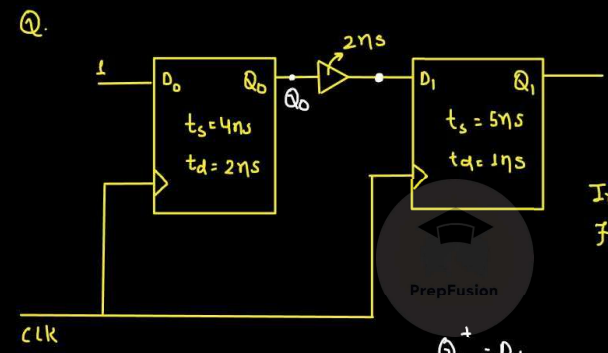
$Q_0^+ = 1$

$Q_1^+ = Q_0$

Q_0Q_1 (initial) = 00

Input at D_0 is constant. Find the minimum Time period of the clock for proper operation.

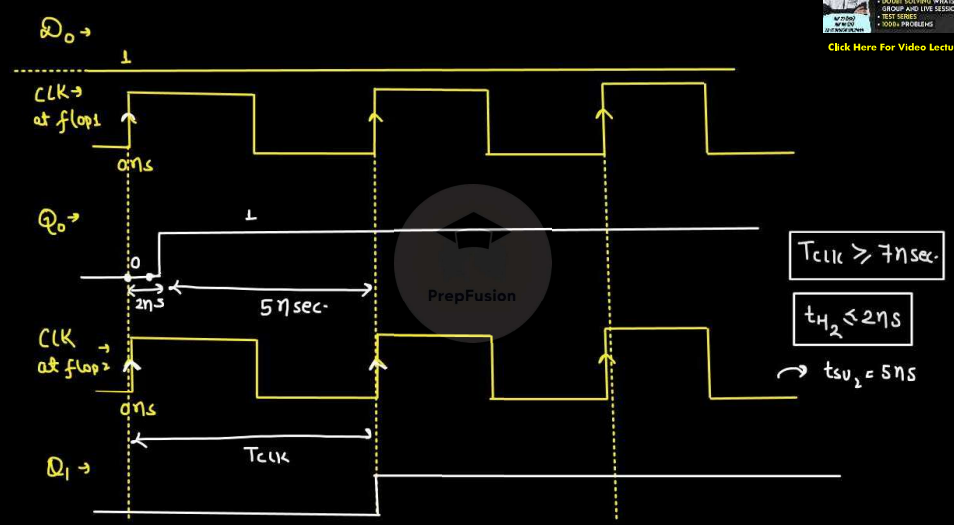
Q.



$Q_1^+ = D_1$

Q_0Q_1 (initial) = 00

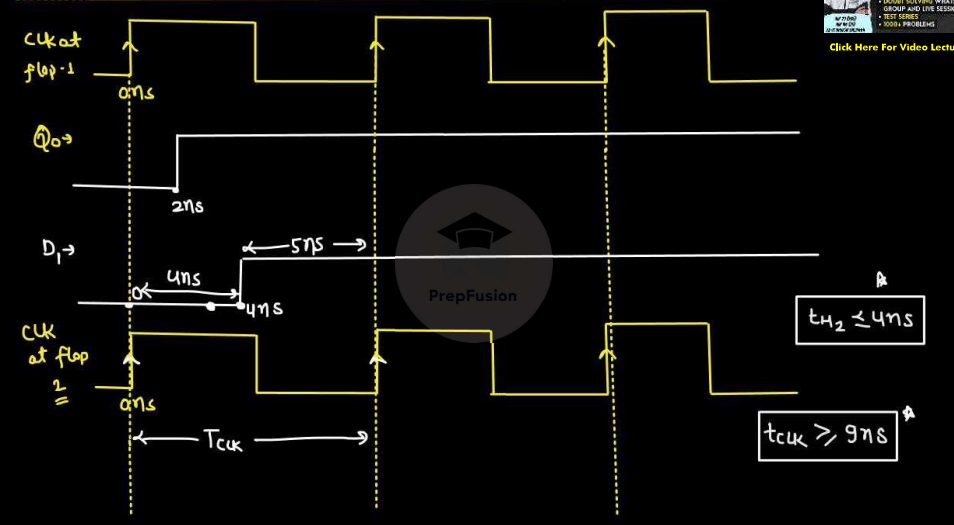
Input at D_0 is constant. Find the minimum Time period of the clock for proper operation.



$T_{CLK} \geq 7ns$ sec.

$t_{H_2} \leq 2ns$

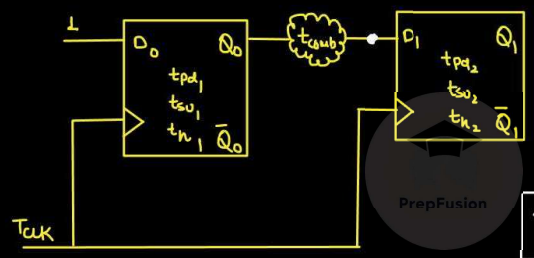
$t_{su_2} = 5ns$



$t_{H_2} \leq 4ns$

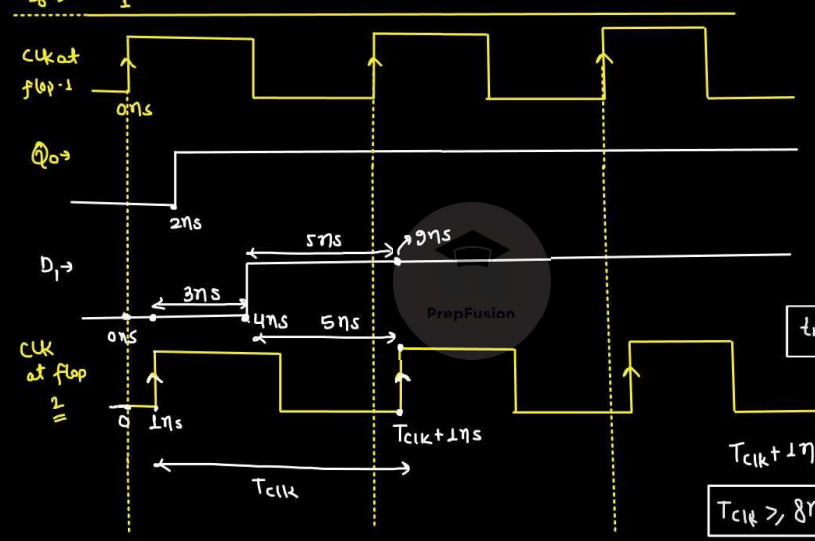
$t_{CLK} \geq 9ns$

Conclusion:-



$$T_{clk} \geq \overbrace{t_{pd_1}}^{t_{cq}} + T_{comb} + t_{su_2}$$

$$t_{H_2} \leq t_{pd_1} + T_{comb}$$

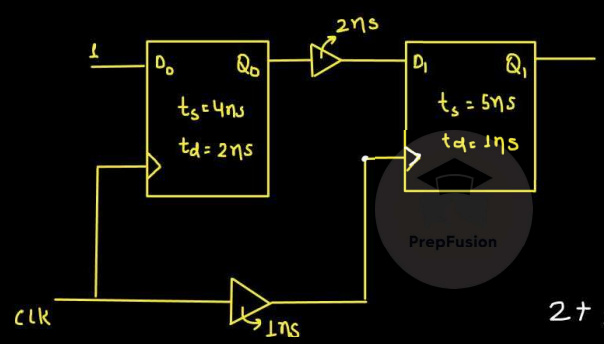


$$t_{H_2} \leq 3ns$$

$$T_{clk}+1ns \geq 9ns$$

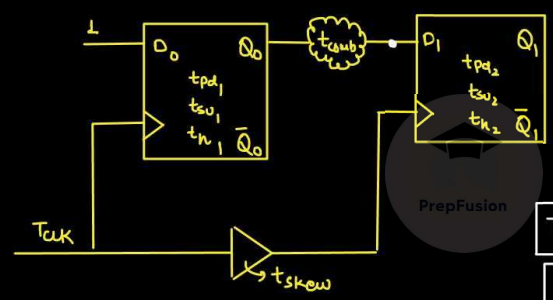
$$T_{clk} \geq 8ns$$

* Introducing clock skew:-



$2 + 2 + 5 - 1$

Conclusion:-



$$T_{clk} \geq t_{pd_1} + t_{comb} + t_{su_2} - t_{skew}$$

$$t_{H_2} \geq t_{pd_1} + t_{comb} - t_{skew}$$

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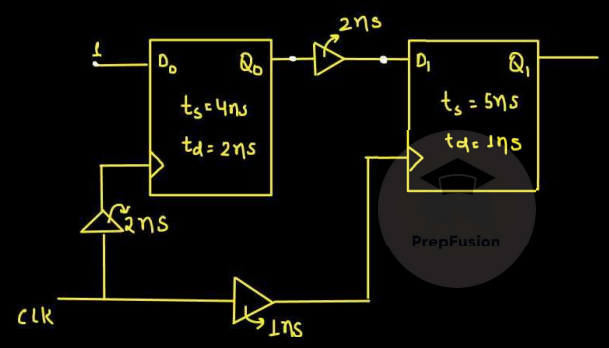
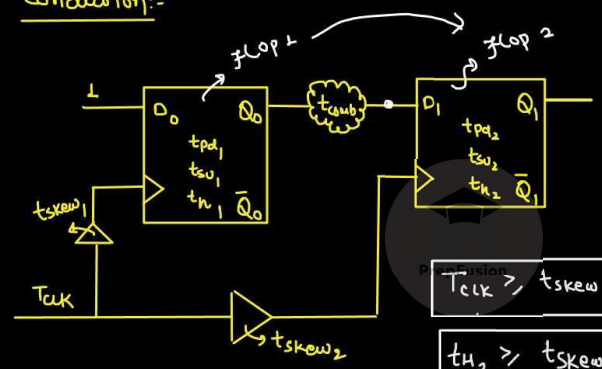
* Introducing clock skew in both of the clocks:-

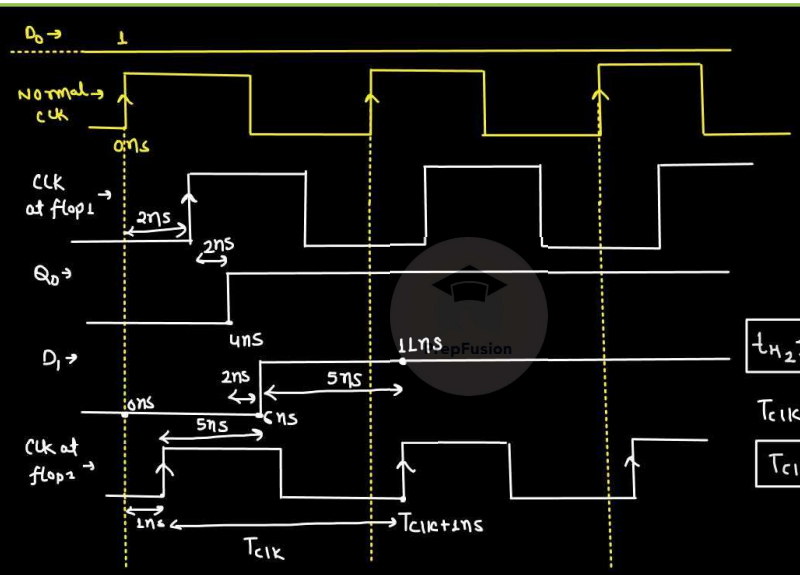


Conclusion:-



$$T_{clk} \geq t_{skew_1} + t_{pd_1} + t_{comb} + t_{su_2} - t_{skew_2}$$

$$t_{h_2} \geq t_{skew_1} + t_{pd_1} + t_{comb} - t_{skew_2}$$

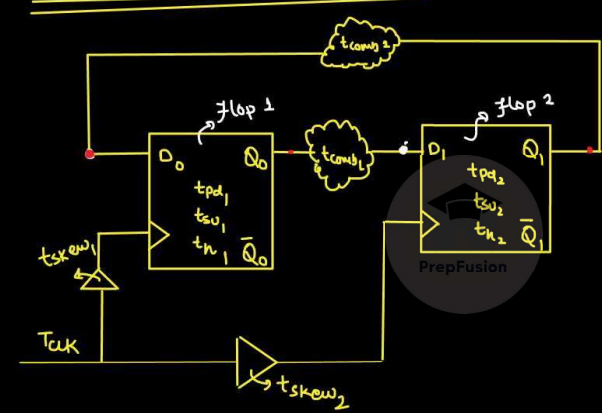


$$t_{h_2} \leq 5ns$$

$$T_{clk} + 1ns \geq 11ns$$

$$T_{clk} \geq 10ns$$

* If both inputs are not fixed:-



(i) Flop 1 to Flop 2

↳ Input at flop 1 is constant [$D_1 \to$ constant]

$$\boxed{T'_{clk} \geq T_{skew_1} + T_{pd_1} + T_{comb_1} + T_{su_2} - T_{skew_2}}$$

$$\to \boxed{t_{H_2} \leq T_{skew_1} + T_{pd_1} + T_{comb_1} - T_{skew_2}}$$

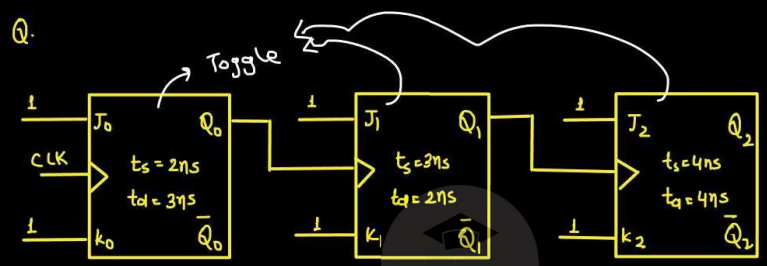
(ii) Flop 2 to Flop 1

↳ Input at flop 2 is constant [$D_2 \to$ constant]

$$\boxed{T''_{clk} \geq T_{skew_2} + T_{pd_2} + T_{comb_2} + t_{su_1} - T_{skew_1}}$$

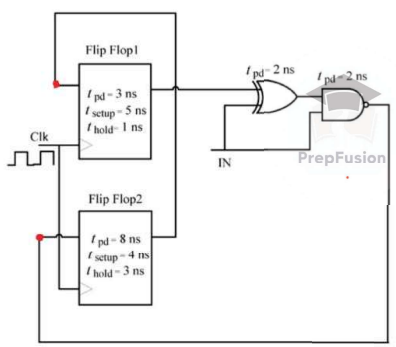
$$\to \boxed{t_{H_1} \leq T_{skew_2} + T_{pd_2} + T_{comb_2} - T_{skew_1}}$$

Q.



For proper opⁿ; maxᵐ allowable clock freq.
[Avoid skipping count]

Q. For the components in the sequential circuit shown below, t_{pd} is the propagation delay, t_{setup} is the setup time, and t_{hold} is the hold time. The maximum clock frequency **(rounded off to the nearest integer)**, at which the given circuit can operate reliably, is _____ MHz.



Flop 1 to Flop 2

$T'_{clk} \geq 3n + 2n + 2n + 4n$

$T'_{clk} \geq 11ns$

Flop 2 to Flop 1

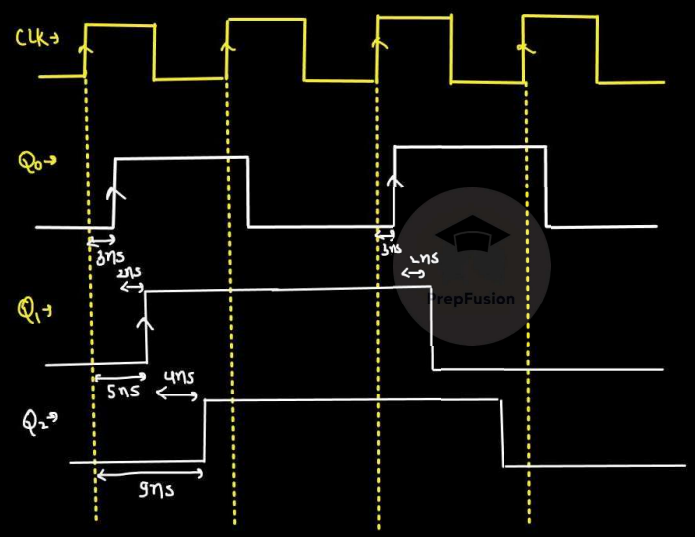
$T''_{clk} \geq 8n + 5n$

$T''_{clk} \geq 13ns$

$\boxed{T_{clk} \geq 13ns}$

$f_{clk} \leq 76.92$ MHz

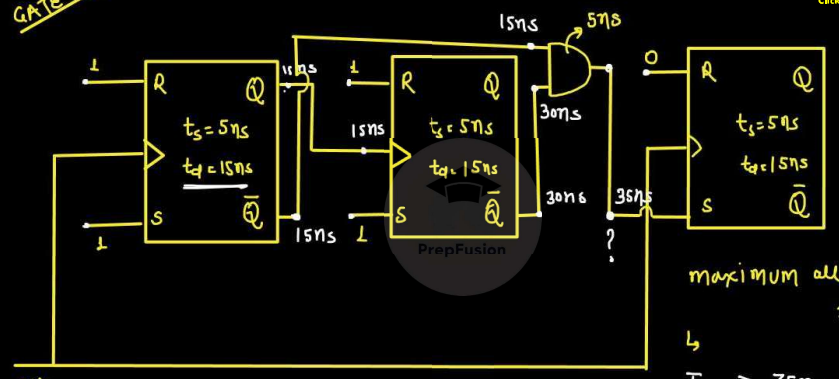
$f_{clk} \leq \frac{1}{13n}$



$T_{clk} \geq 3n + 2n + 4n$

$\boxed{T_{clk} \geq 9ns}$

Q. GATE 2004



maximum allowable freq?

↳

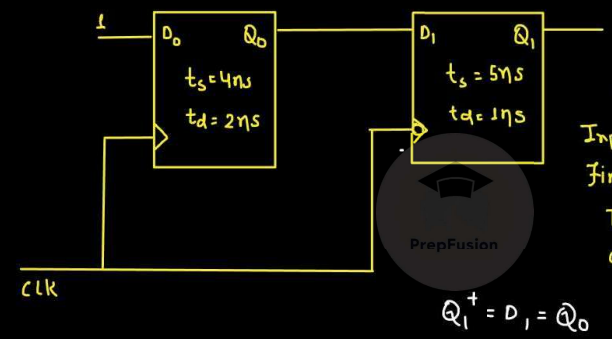
$T_{CLK} \geq 35n + 5n$

$T_{CLK} \geq 40n$

$f_{CLK} \leq \frac{1}{40n}$

$\boxed{f_{CLK} \leq 25MHz}$

Q.

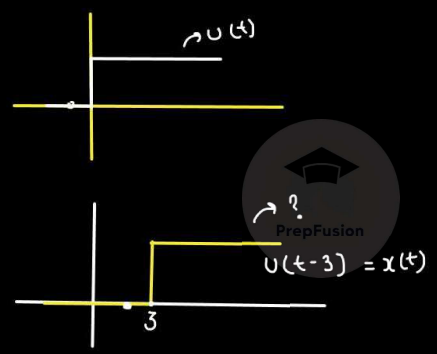


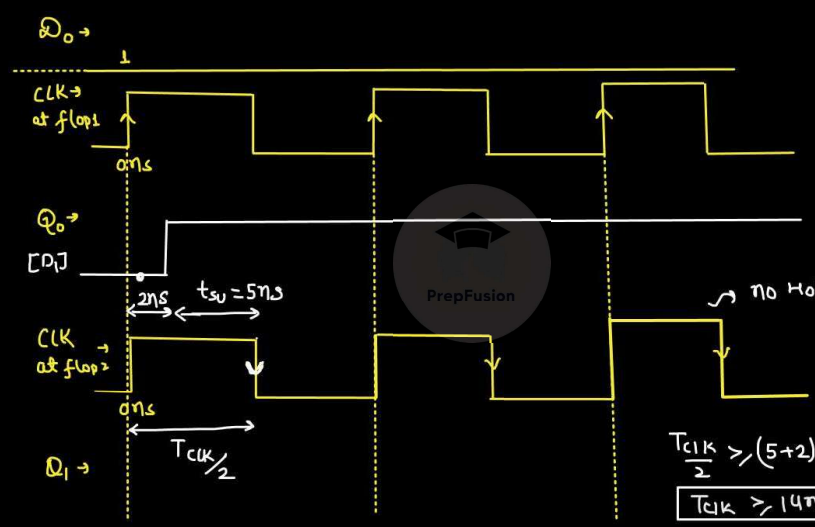
$Q_0 Q_1$ (initial) = 00

Input at D_0 is constant. Find the minimum Time period of the clock for proper operation.

$Q_1^+ = D_1 = \bar{Q}_0$

Small Conceptual understanding:-





no Hold violation at all

$\frac{T_{CLK}}{2} \geq (5+2)ns$

$\boxed{T_{CLK} \geq 14ns}$

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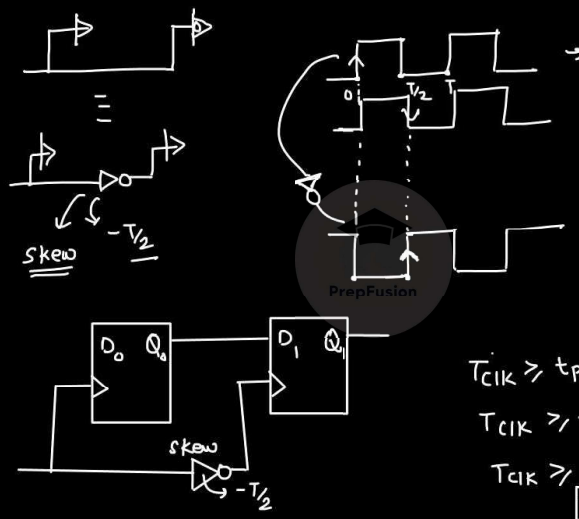

→ Pos edge Trigger

→ Neg edge Triggered

$3T_{clk}/2 \geq 7$ (won't consider)

$T_{clk} \geq 14/3$

$T_{clk} \geq t_{pd_1} + t_{su_2} - \left(-\frac{T_{clk}}{2}\right)$

$T_{clk} \geq t_{pd_1} + t_{su_2} + T_{clk}/2$

$T_{clk} \geq 2[2n + 5n]$

$T_{clk} \geq 14ns$

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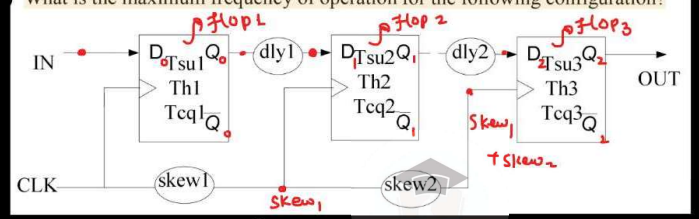
From digi-Qs →

What is the maximum frequency of operation for the following configuration?



Flop 1 to Flop 2

$T'_{clk} \geq T_{cq_1} + T_{dly_1} + T_{su_2} - T_{skew_1}$

$T_{H_2} \leq T_{cq_1} + T_{dly_1} - T_{skew_1}$

Flop 2 to Flop 3

$T''_{clk} \geq T_{skew_1} + T_{cq_2} + T_{dly_2} + T_{su_3} - (T_{skew_1} + T_{skew_2})$

$T''_{clk} \geq T_{cq_2} + T_{dly_2} + T_{su_3} - T_{skew_2}$

$T_{H_3} \leq T_{cq_2} + T_{dly_2} - T_{skew_2}$

$[T_{clk}]_{min} = \max[T'_{clk}, T''_{clk}]$

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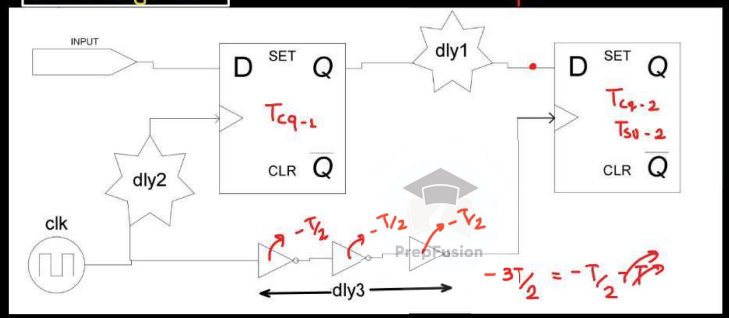
From digi-Qs :-



$(T_{clk})_{min} = ?$

$T_{clk} \geq T_{dly_2} + T_{cq-1} + T_{dly-1} + T_{su-2} - \left(-\frac{T_{clk}}{2} + T_{dly_3}\right)$

$T_{clk} \geq 2[T_{dly_2} + T_{cq-1} + T_{dly-1} + T_{su-2} - T_{dly_3}]$

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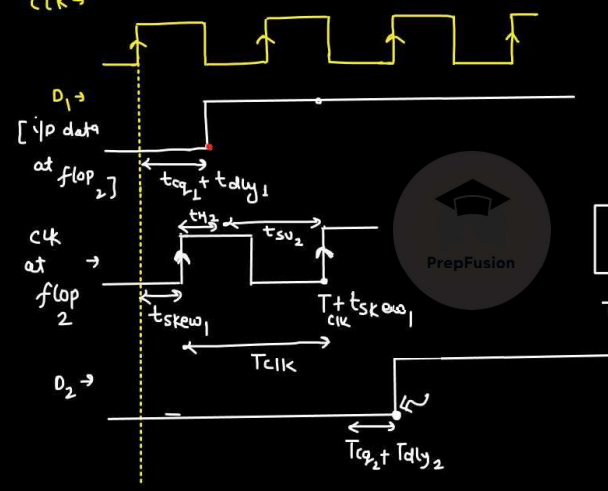

$t_{H_2} \leq t_{cq_1} + t_{dly_1} - t_{skew_1}$

$T_{clk} + t_{skew_1} \geq t_{cq_1} + t_{dly_1} + t_{su_2}$

$T_{clk} \geq t_{cq_1} + t_{dly_1} + t_{su_2} - t_{skew_1}$

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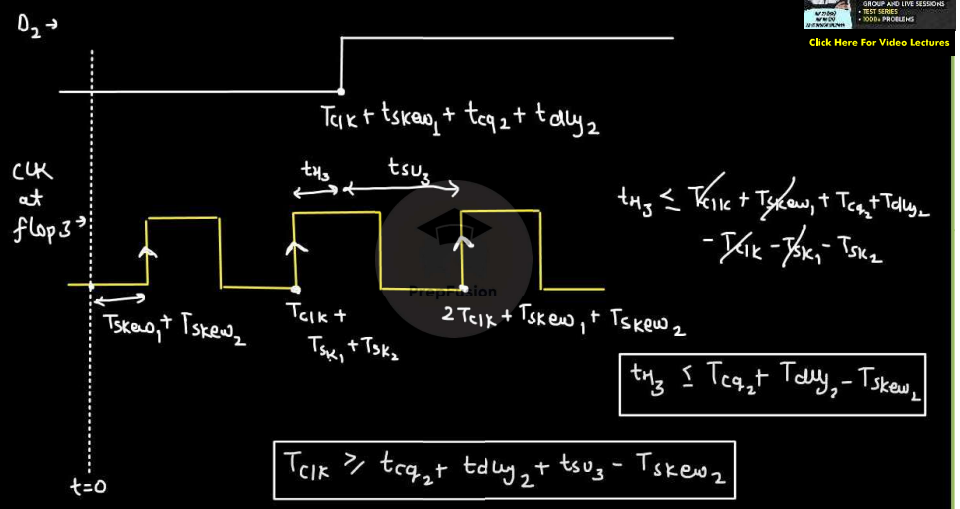

$$t_{H_3} \le \cancel{T_{clk}} + \cancel{T_{skew_1}} + T_{cq_2} + T_{dly_2} - \cancel{T_{clk}} - \cancel{T_{sk_1}} - T_{sk_2}$$

$$\boxed{t_{H_3} \le T_{cq_2} + T_{dly_2} - T_{skew_2}}$$

$$\boxed{T_{clk} \geq t_{cq_2} + t_{dly_2} + t_{su_3} - T_{skew_2}}$$

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(a) $T_{clk} \geq 6\eta + 5\eta + 3\eta + 2\eta$

$T_{clk} \geq 16\eta$

$f_{clk} \le \frac{1}{16\eta}$

$f_{clk} \le 62.5\,MHz$ $\boxed{(f_{clk})_{max} = 62.5\,MHz}$

(b) I want to operate it over 62.5 MHz. What can I do?

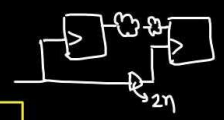
<u>Method-1</u>:- Introduce a clock skew [let it be 2η]

$T_{clk} \geq 6\eta + 5\eta + 3\eta + 2\eta - 2\eta$

$T_{clk} \geq 14\eta$

$f_{clk} \le \frac{1}{14\eta} \Rightarrow f_{clk} \le 71.42\,MHz$

$\boxed{(f_{clk})_{max} = 71.42\,MHz}$



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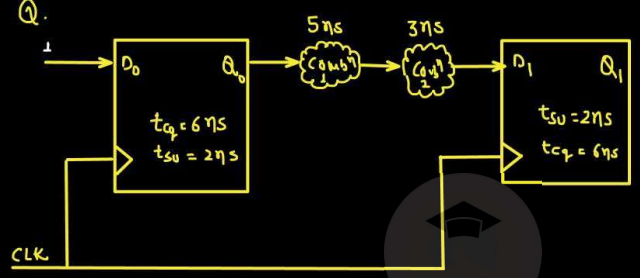
Q.



(a) maximum clock freq.?

(b) If you want to operate the given circuit at a higher freq. than the max^m freq. you calculated, then what modification you will do?

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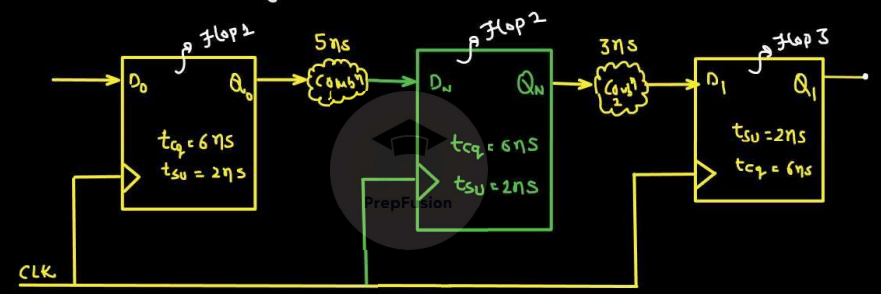
<u>Method-2 (Pipelining)</u>:-

Introducing a new flip-flop in between



Flop 1 to Flop 2

$T_{clk} \geq 6 + 5 + 2$

$T_{clk} \geq 13\eta s$

Flop 2 to Flop 3

$T_{clk} \geq 6\eta + 3\eta + 2\eta$

$T_{clk} \geq 11\eta$

$\Rightarrow$ $T_{clk} \geq 13\eta s$

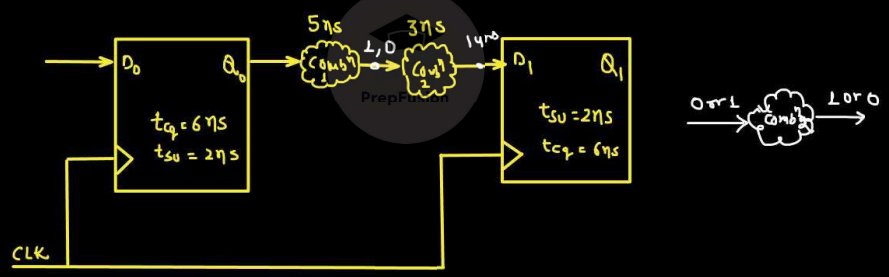
$\boxed{f_{clk} \le 76.92\,MHz}$

Drawback of pipelining :-

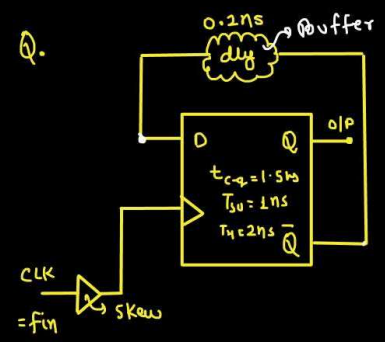
↳ Increased latency

↳ Increased Area

Method - 3 :-



Q.



(i) $Q^+ = \bar{Q}$ → Toggle

$f_o = f_{in}/2$

(ii) $t_H \leq 1.5ns + 0.2ns$

But $t_H = 2ns$ (Given) ⇒ Hold violation

(iii) if dly goes to 0.5

$t_H \leq 1.5ns + 0.5ns$

$\boxed{t_H \leq 2ns}$

(i) o/p freq. = ?

(ii) minimum Time period required

(iii) Hold violation present ?

(iv) Any effect of skew ? → NO

(v) Suggestion to avoid hold violation ?

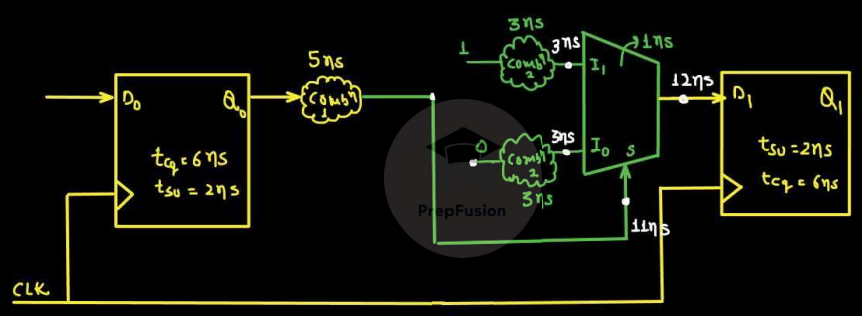
↳ Increase dly delay.

(ii) Flop 1 to Flop 1

$T_{clk} \geq T_{skew} + 1.5ns + 0.2ns + 1ns - T_{skew}$

$\boxed{T_{clk} \geq 2.7ns}$

Since, I/p to comb^n 2 could be either "0" or "1".

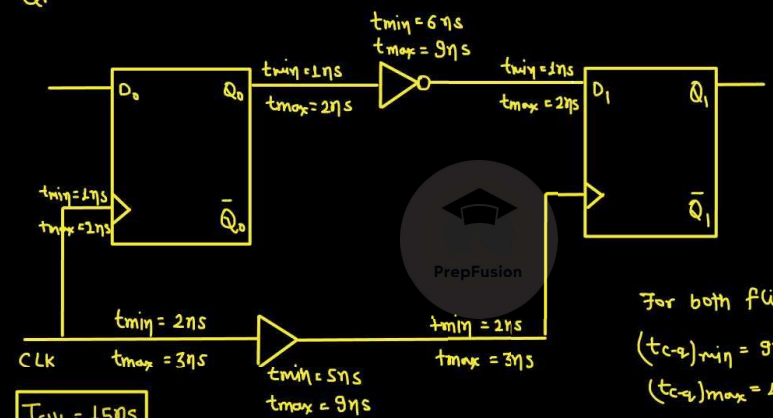


$T_{clk} \geq 6 + 5 + 1 + 2$

$T_{clk} \geq 14ns$

$\boxed{f_{clk} \leq 71.42MHz}$

Q.



$\boxed{T_{clk} = 15ns}$

(i) Analyze if there is any setup or hold violation. Find setup slack.

For both flip-flops :-

$(t_{c-q})_{min} = 9ns$

$(t_{c-q})_{max} = 11ns$

$t_{hold} = 2ns$

$t_{setup} = 4ns$

$$T_{CLK} \geq \underbrace{\underbrace{2n + 11n + 2n + 9n + 2n}_{max} + \overset{\text{setup time}}{4n} - \underbrace{(2n + 5n + 2n)}_{min}}_{maximum}$$

$T_{CLK} \geq 21ns$ But $T_{CLK} = 15ns \rightarrow$ Given

$(T_{CLK})_{min} = 21ns$

↳ setup violation

↓

setup time should be decreased by 6ns

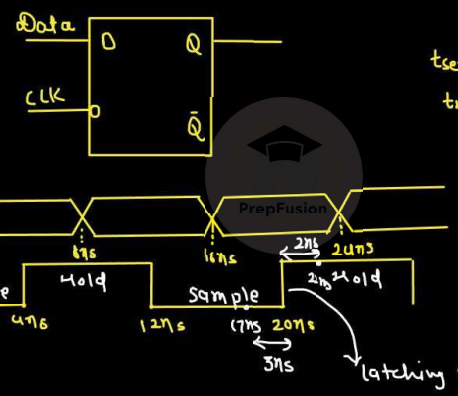
setup slack = $-6ns$

$t_{H_2} \leq 1n + 9n + 1n + 6n + 1n - (3n + 9n + 3n)$ {min}

$\boxed{t_{H_2} \leq 3ns}$ $t_{H_2} = 2ns$ ⇝ given ⇒ no Hold violation

Q. what do you mean by setup time and hold time of a latch?

↳



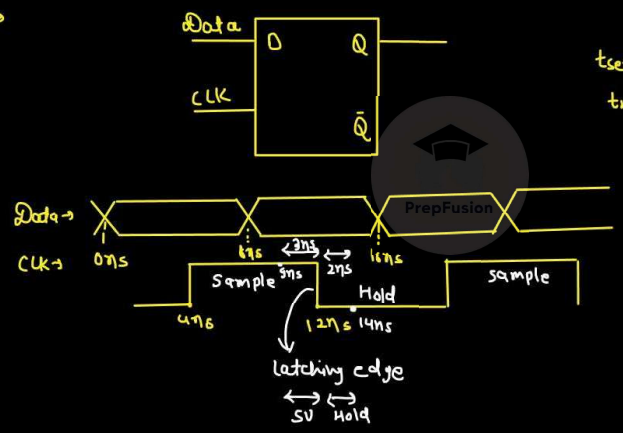
$t_{setup} = 3ns$

$t_{hold} = 2ns$

⇝ NO setup or Hold violation

Q. what do you mean by setup time and hold time of a latch?

↳



$t_{setup} = 3ns$

$t_{hold} = 2ns$

⇝ no setup or Hold violation

Definition:-

↳ Setup time (Latch):- minimum time i/p needs to be stable prior to the latching edge.

↳ Hold time (Latch):- minimum time i/p needs to be stable after the latching edge.

MSQ

Q. How to fix setup violation?

(a) Speed up the clock → $f(\uparrow) \Rightarrow T_{clk}(\downarrow)$

(b) Speed up the data → decreasing data delay

(c) Increase data delay

(d) Increase clock period

PrepFusion

$T_{clk} \geq T_D + T_{setup}$ (Data delay)

$$\boxed{T_{setup} \leq T_{clk} - T_D}$$

$T_s \leq 5ns$

[b, d] $T_s \leq 6ns$

[a, c] $T_s \leq 4ns$

$\boxed{5ns}$

5.5ns

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MSQ

Q. How to fix Hold violation?

(a) Speed up the clock

(b) Speed up the data → decreasing T_D

(c) Increase data delay

(d) Reduce clock skew

PrepFusion

$t_H \leq T_D - T_{skew}$

$t_H \leq 5ns$

$t_H \leq 6ns$ [c, d]

$t_H \leq 4ns$ [a, b]

$\boxed{5ns}$

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